



Salem Stays: What Drives Airbnb Ratings and Pricing in Oregon's Capital?

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Main Research Question

What listing and host characteristics predict an overall review score and pricing in Salem's Airbnb market?

Sub Questions

1

Do certain amenities make the Airbnb more popular/more expensive?

2

Are there certain “key” words mentioned in the reviews, and does it correlate to higher reviews?

3

Does proximity to Salem landmarks affect pricing or ratings?

4

Is Salem’s Airbnb market growing, and are there seasonal patterns?

5

Do host characteristics influence ratings?



Datasets

All collected from: Inside Airbnb and renamed

salem_detailed_listings.csv

Detailed columns including
ratings

Price data is all null
85 columns, 280 rows

salem_listings.csv

Used to get price
values

Merged using listing_id
18 columns, 352 rows

salem_reviews.csv

21,000+ reviews
Includes listing_id for
merging
Will be used for feature
engineering



Data Cleaning & Sanity Check

- After merging datasets, there were some nulls

```
Nulls in price: 50  
Missing IDs not in summary file: 34  
Missing IDs that ARE in summary but price is null: 16
```

```
Room types of listings with missing price:
```

```
room_type  
Entire home/apt    38  
Private room      12  
Name: count, dtype: int64
```

```
Do they have ratings?  
37
```

→ Imputed the median price per room type for the null values in price

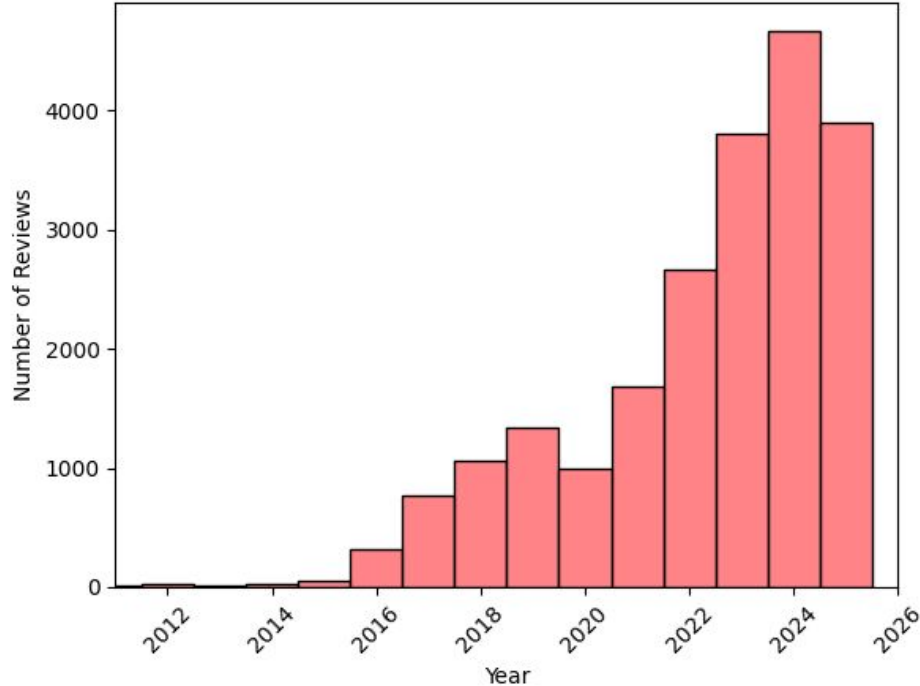
- Listings shape: (280, 85)
Reviews shape: (21319, 6)

```
Price range: $30 – $1000  
Listings with ratings: 246  
Reviews date range: 2011-08-29 – 2025-09-24
```

* 85 unique columns after merging the 2 datasets

Trends Overtime

Salem Airbnb Reviews 2011-2025

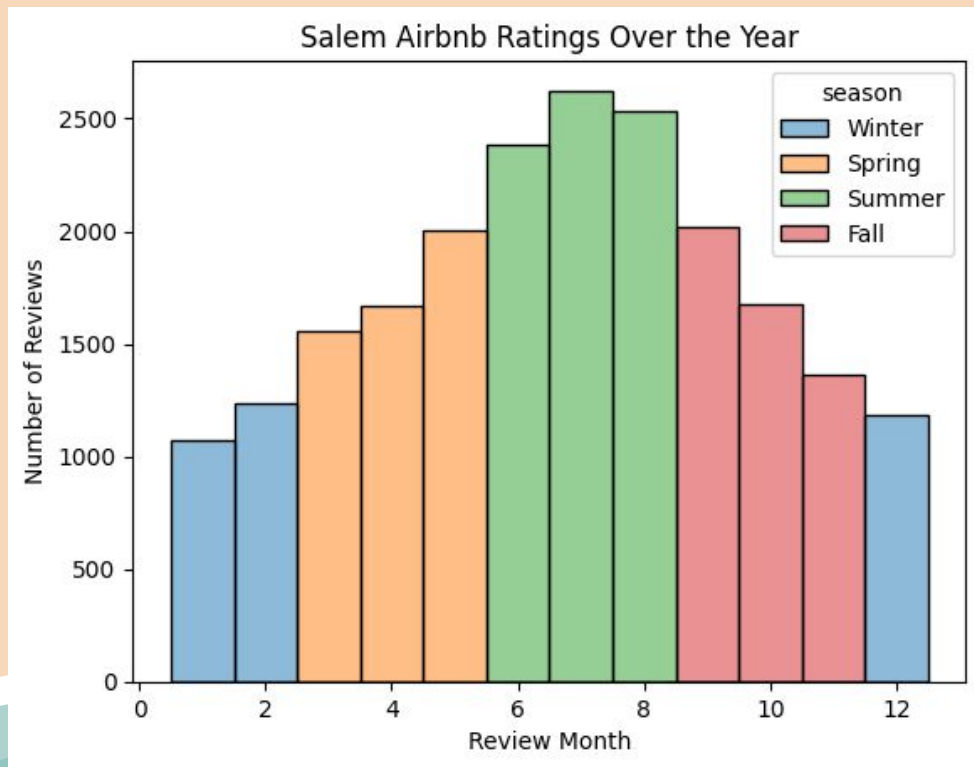


Activity has increased significantly between 2011-2025!

What this could mean:

- Growing need/market
- For hosts, growing competition and greater need to be strategic in how you list/present your airbnb

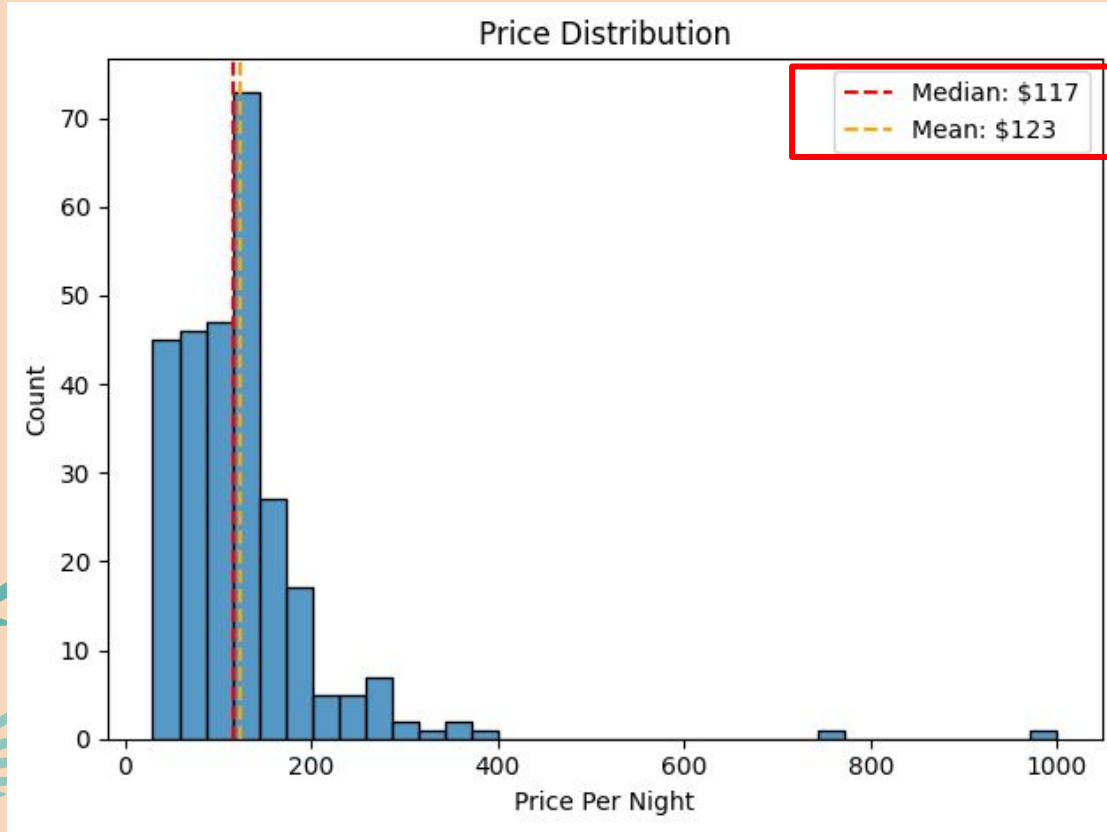
Trends Overtime



Reviews/Activity peak in the summer months!

What this could mean:

- Summer months are a popular time for booking
- Taking advantage of this time with how you list amenities (highlighting outdoor/summer related amenities)

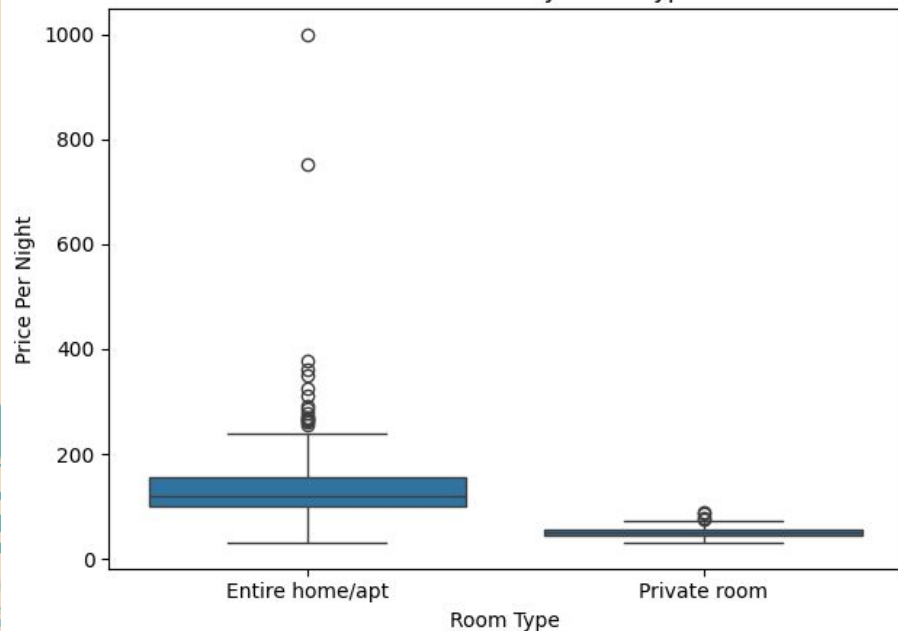


Price Distribution

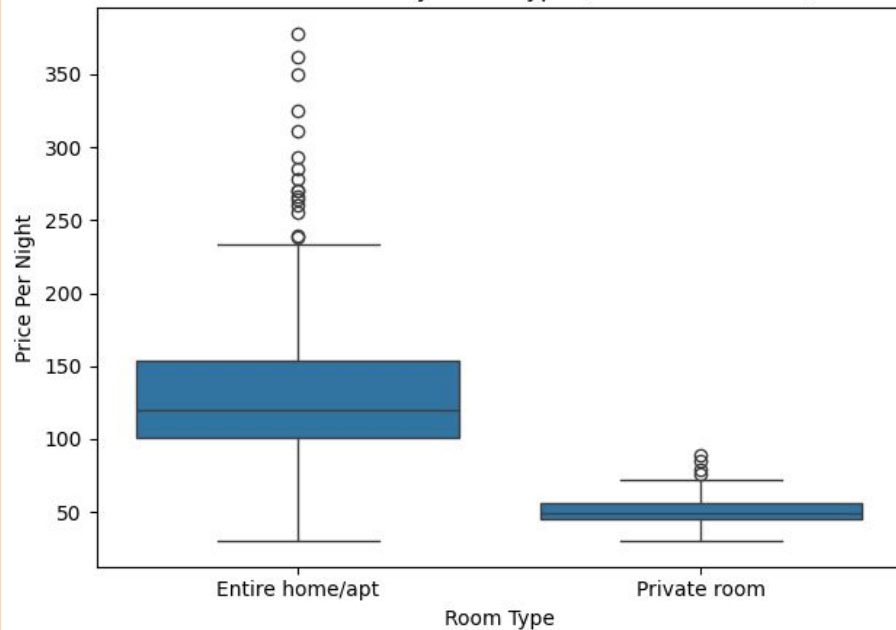
- Outliers are slightly skewing the mean
- Most properties fall between \$50 - 200

Price Distribution

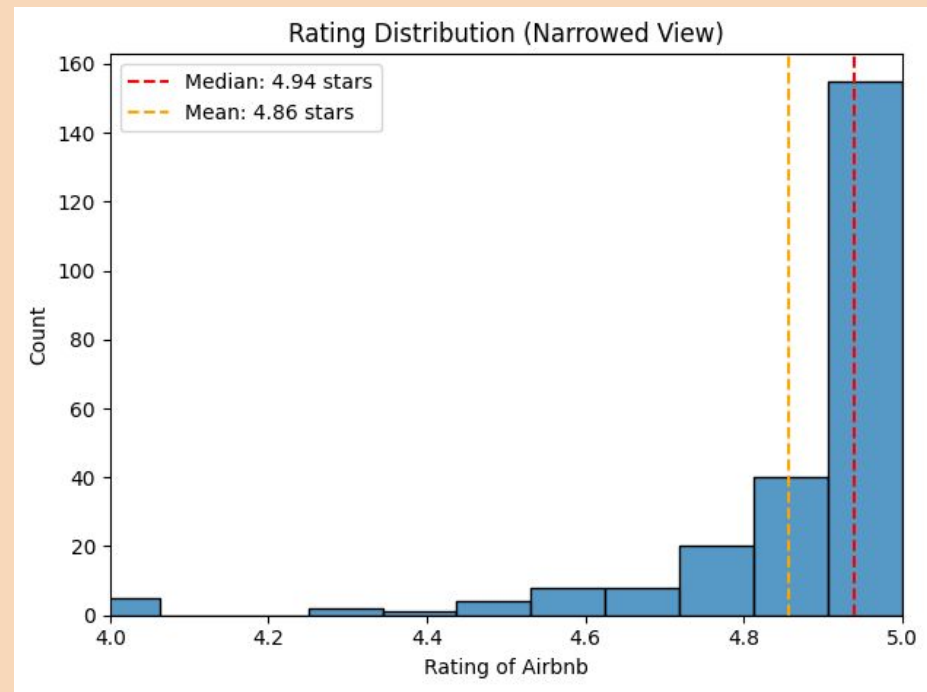
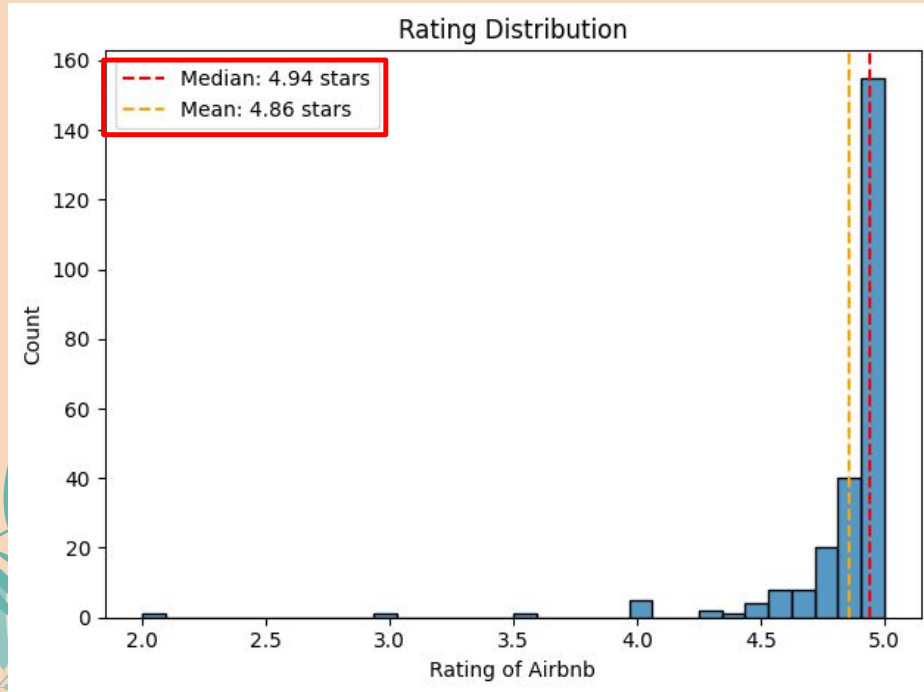
Price Distribution by Room Type



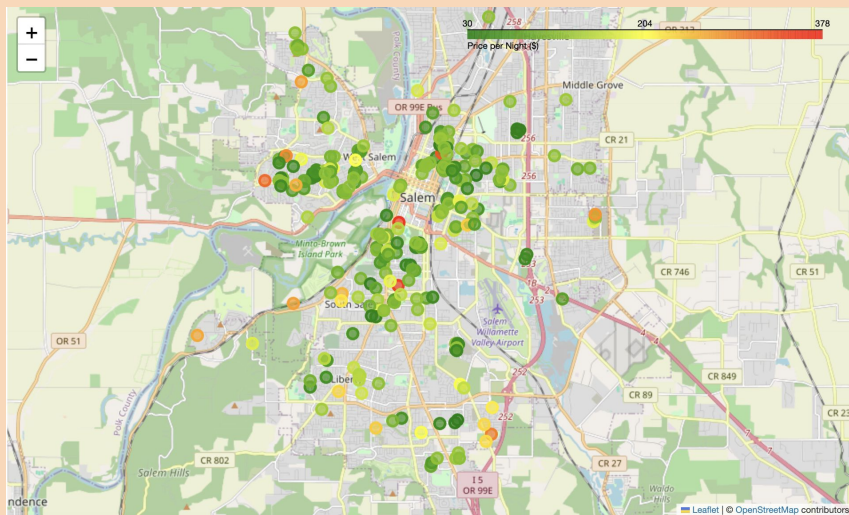
Price Distribution by Room Type (Outliers Removed)



Rating Distribution

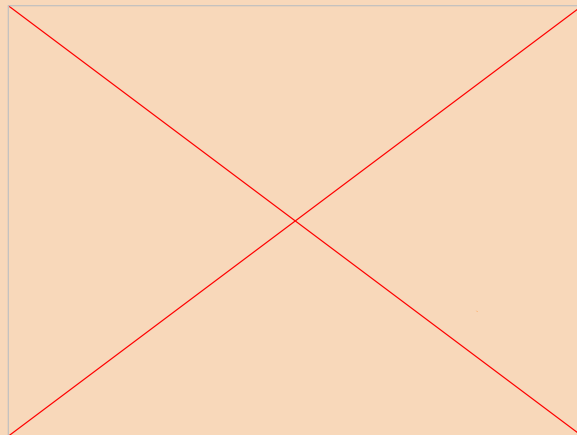


Geo Features

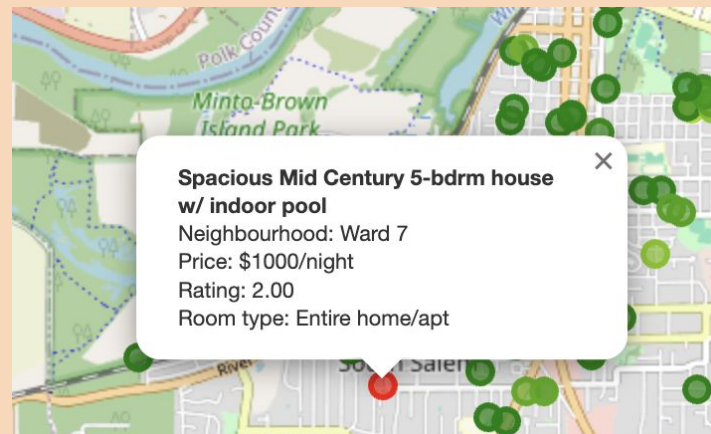
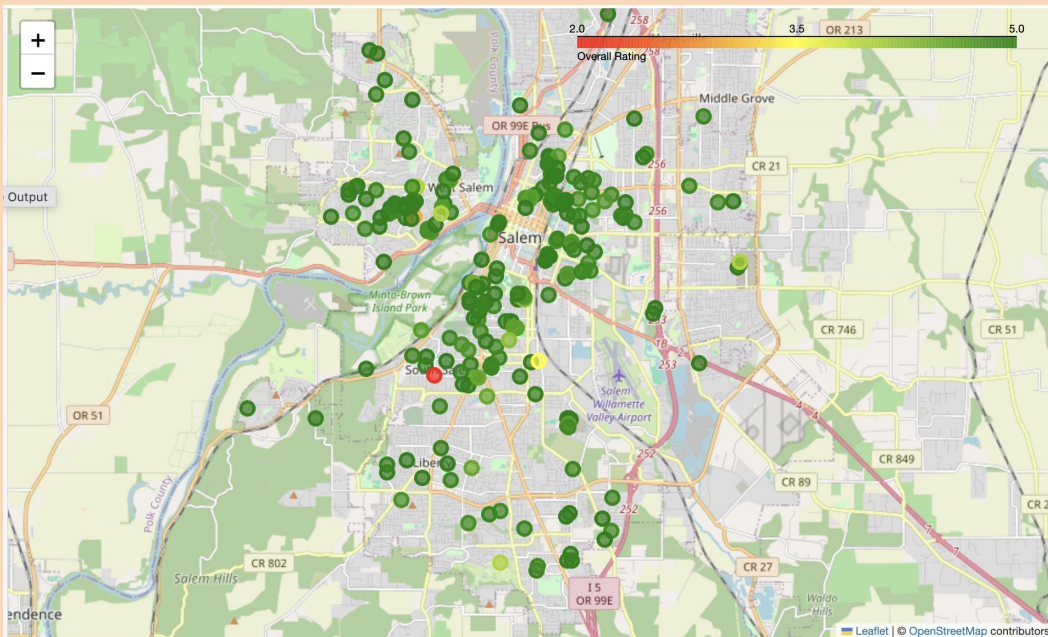


Map 1: Color by Price (outliers removed)

- West Salem is pretty pricey (the hills)
- Historical homes + Outskirts of Salem for Willamette Valley, Wineries, etc.
- Landmarks mentioned frequently: Mill Creek, Hospital, Willamette, Capitol, Riverfront, Downtown, Bush Park



Geo Features



Map 2: Color by Rating

- Overall solid ratings (no relation to landmarks)
- Lowest rating is the \$1000 house lol

* Map is interactive in Jupyter notebook, using folium

Feature Engineering & Models



Feature Engineering

Superhosts:

- A superhost is a highly rated, experienced host who consistently provides reliable stays, excellent communication, and outstanding guest experiences.
- Data mapping:
 - T or F → **0** or **1**

Other review categories:

- | | |
|--------------------|-----------------|
| • Accuracy | • Communication |
| • Cleanliness | • Location |
| • Check-In Process | • Value |

Neighborhood:

- This variable is broken up into the wards/neighborhoods of Salem
- Data mapping:
 - Ward 1 → **0**
 - Ward 2 → **1**
 - Ward 3 → **2**
 - etc...

Host Response Time:

- This data is in 4 categories such as “Within an hour”, “Within a day”, etc.
- Data mapping:
 - Within an hour → **4**
 - Within a few hours → **3**
 - Within a day → **2**
 - A few days or more → **1**

Linear Regression: Host Characteristics

Superhost:

- Intercept: 4.87
- Coefficient: 0.05

Interpretation:

- For non-superhosts, we expect a 4.87
- We expect rating will increase by 0.05 if superhost



Host Response Time:

- Intercept: 4.94
- Coefficient: -0.006

Interpretation:

- This means that the faster a host responds, we expect a decrease in score



Multiple Linear Regression: All Review Categories

Review rating categories: Overall, accuracy, cleanliness, check-in process, communication, location, value

Multiple Linear Regression Results:

Statistic	Result
Intercept	-1.084
Accuracy Coefficient	0.463
Cleanliness Coefficient	0.264
Check-In Coefficient	0.032
Communication Coefficient	0.298
Location Coefficient	0.051
Value Coefficient	0.107

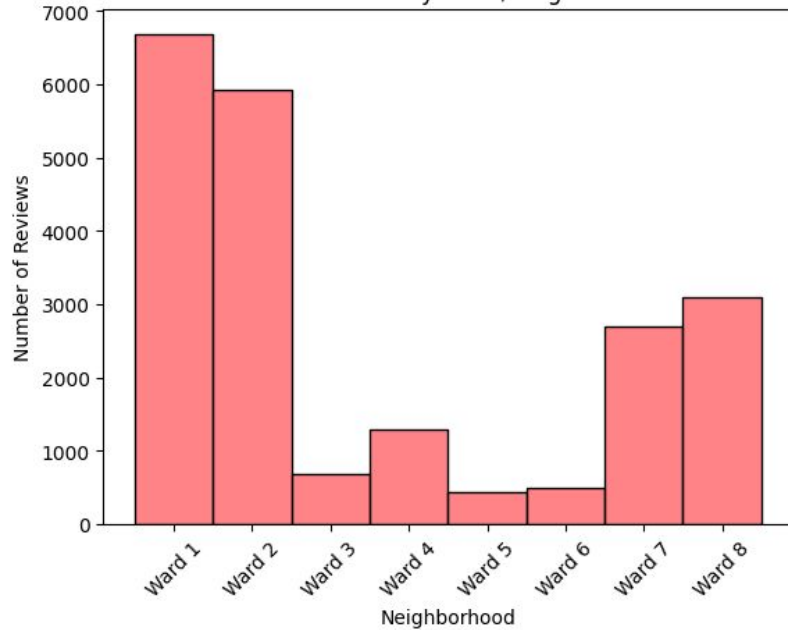
Key Takeaways:

- Location and Check-In Process rating have the lowest impact on the the overall rating.
- Accuracy has the highest impact on the overall rating, followed by communication and cleanliness



Neighborhoods

Salem Reviews by Ward/Neighborhood

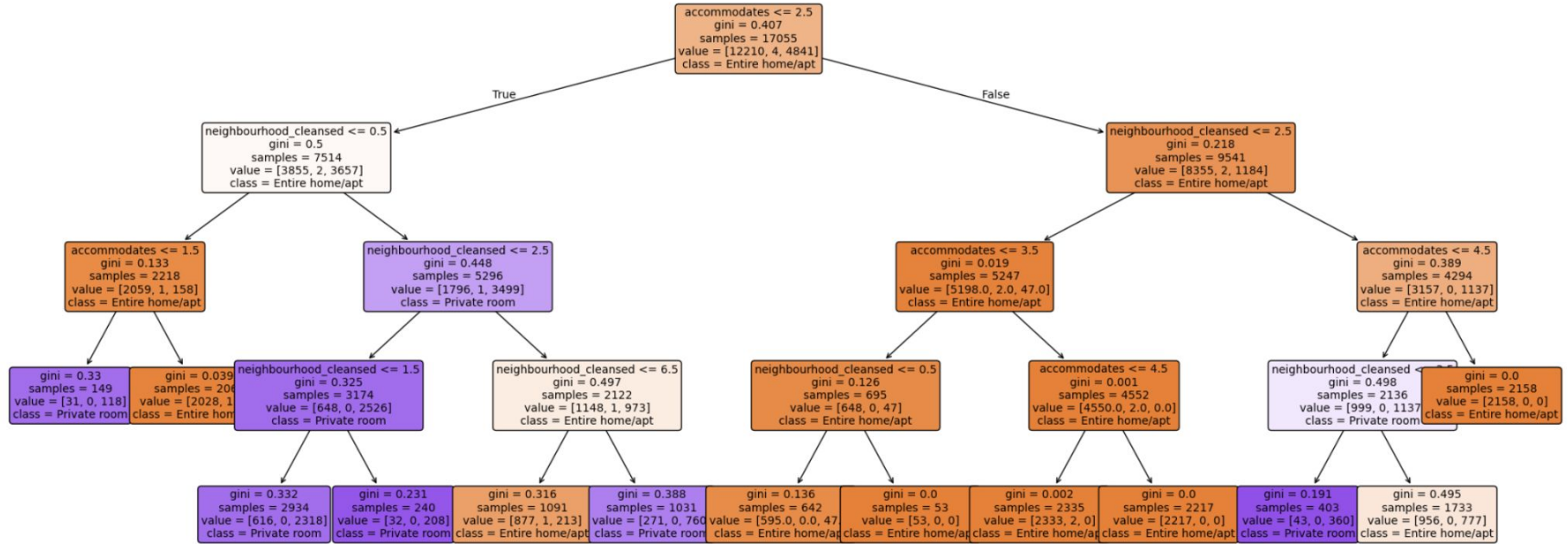


What we are seeing from a neighborhood perspective:

- **Ward 1** - Downtown/Central
- **Ward 2** - Southeast
- **Ward 3** - South/South East
- **Ward 4** - South Gateway
- **Ward 5** - North Salem
- **Ward 6** - East Salem
- **Ward 7** - South
- **Ward 8** - West Salem

Classification - Room Type

Decision Tree - Room Type Prediction



Takeaways: Predicting room type based on the

- Most single rooms and low number of guests
- Lower accommodation number often single rooms

Classification of Ward - KNN

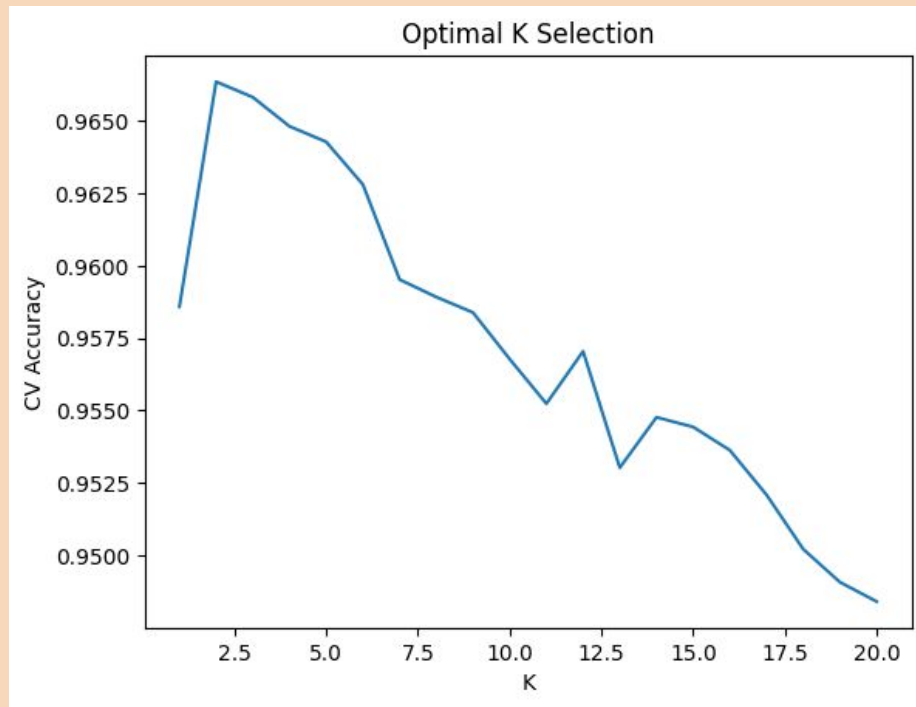
FEATURES: Price, Availability_365,
Minimum Nights

TARGET: Ward/Neighbourhood

Imputed NA's in Price with Median
Price

Train/Test Split - Stratified by
Neighborhood

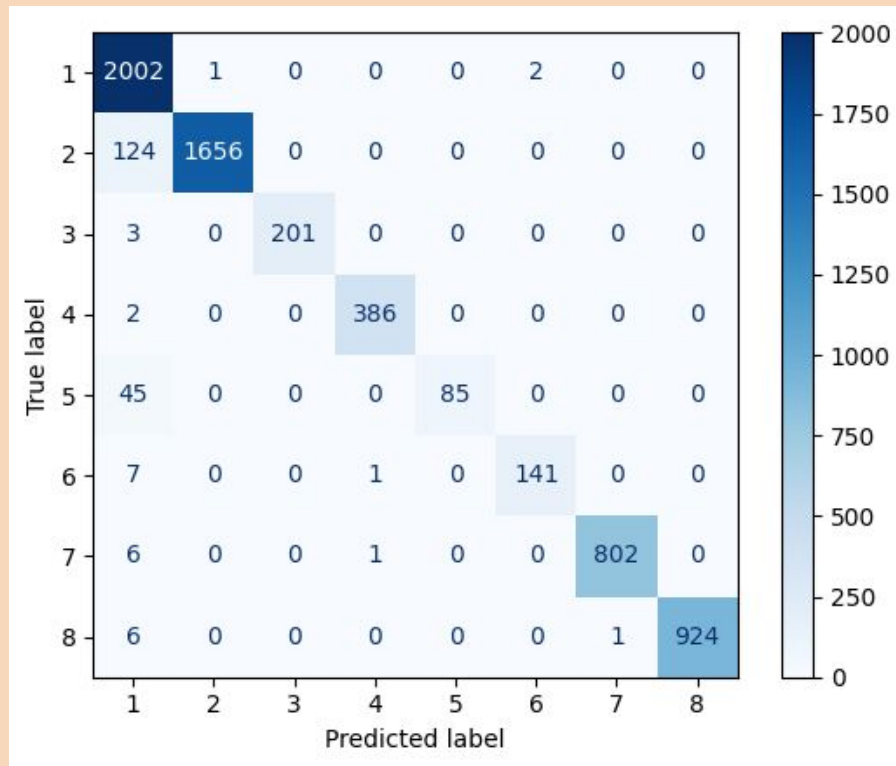
K = 2



Classification of Ward - KNN

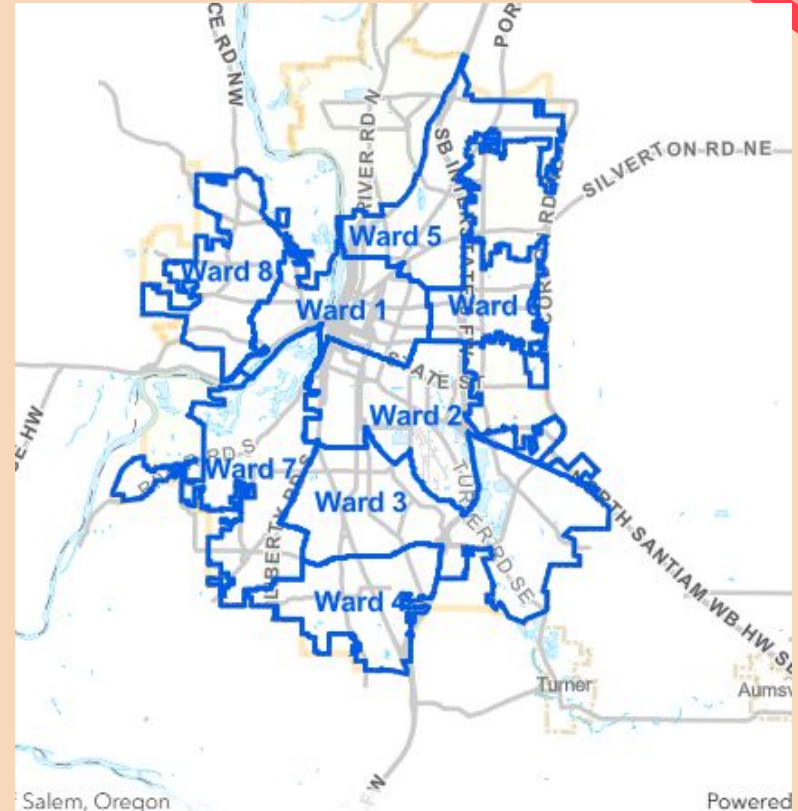
Accuracy	0.9689
Precision	0.9715
Recall	0.9689
F1	0.9685

- WARD 1 & 2 likely have high misclassification
- WARD 5 and WARD 1 also have high misclassification
 - Smaller Class



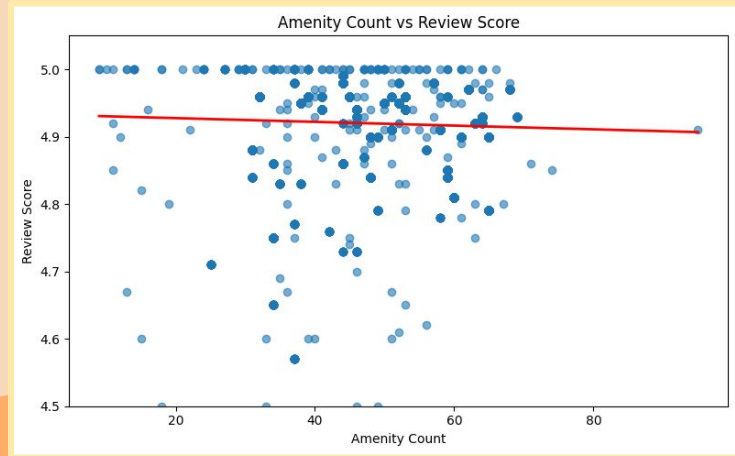
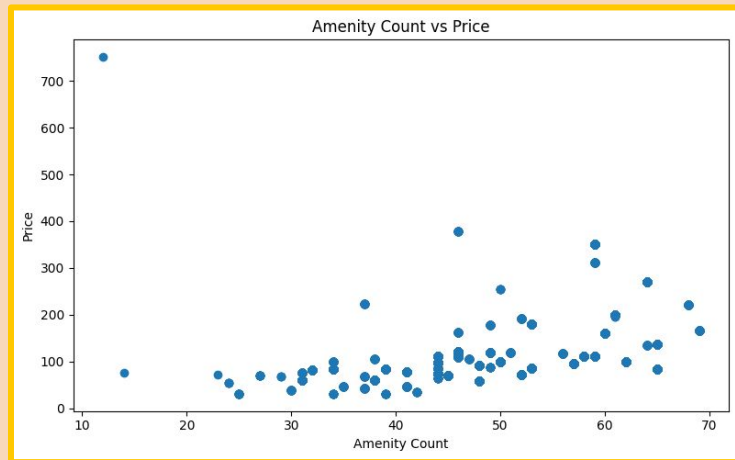
Classification of Ward - KNN

- WARDS 1, 2 and 5 are next to each other
 - **Ward 1** - Downtown/Central
 - **Ward 2** - Southeast
 - **Ward 3** - South/South East
 - **Ward 4** - South Gateway
 - **Ward 5** - North Salem
 - **Ward 6** - East Salem
 - **Ward 7** - South
 - **Ward 8** - West Salem

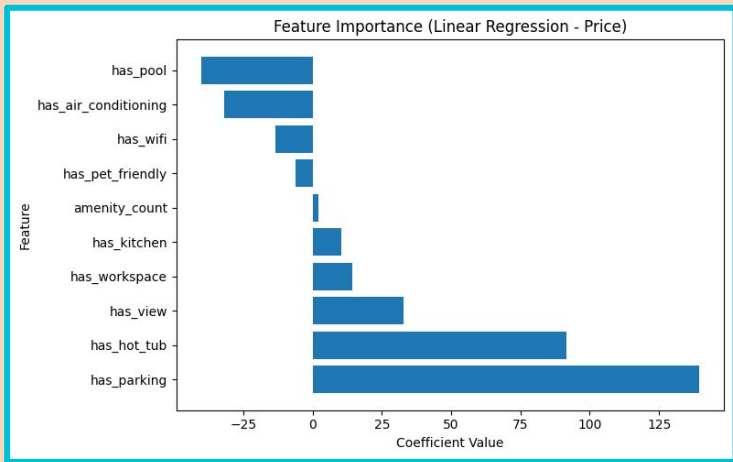


Amenities - Key Findings

- Listings with more amenities tend to have higher prices
- Some amenities have a stronger impact:
 - Parking
 - Hot tub
 - Views
- Basic amenities like wifi or air conditioning
 - Do not significantly increase price
- Amenities do NOT strongly affect review scores
- Ratings are already high across most listings
- This suggests:
 - Amenities influence price
 - But not overall satisfaction as much

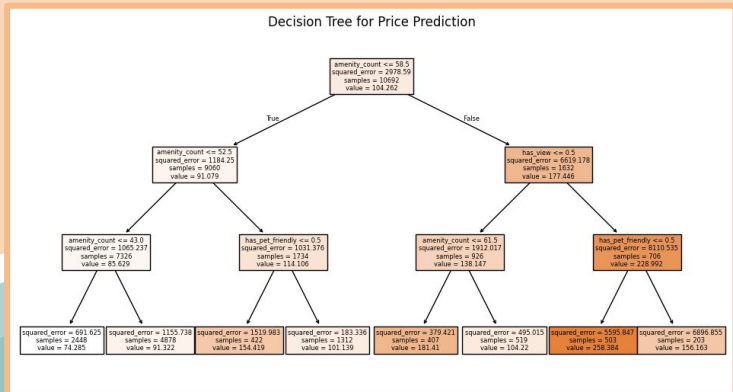


Amenities - Modeling Results



- Linear Regression:
 - $R^2 \approx 0.43$
 - Assumes a linear relationship
 - Each amenity adds or subtracts a fixed amount to price
 - Easier to interpret, but more limited

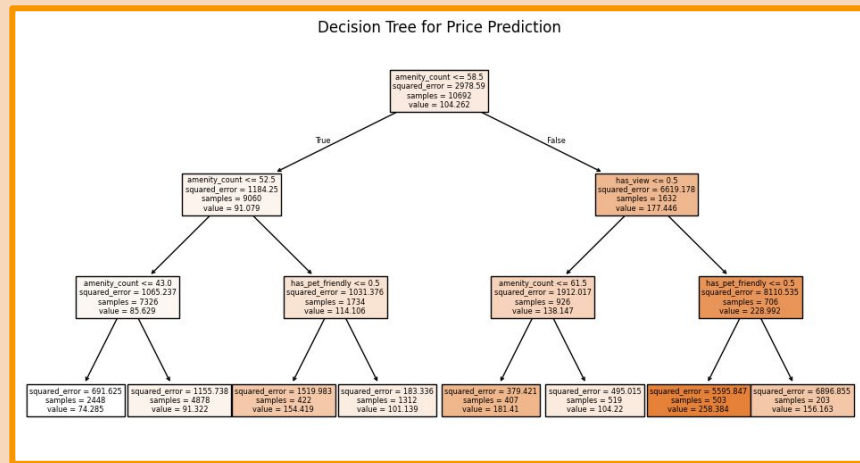
- Decision Tree:
 - $R^2 \approx 0.60$ (better performance)
 - Splits the data into groups based on features



- How the tree works:
 - Starts with all listings at the top
 - Makes splits like:
 - “amenity_count > 58”
 - “has_view > 1”
 - Each split creates more specific groups
 - Ends with a predicted price for each group
- This shows that price depends on combinations of amenities

Amenities - Final Takeaway + Decision Tree Insight

- The decision tree shows how the model makes price predictions step-by-step
- The first split is based on `amenity_count`
 - This means total amenities is the most important factor
- Listings with fewer amenities:
 - Fall into lower price ranges (~ \$70 - \$90)
- Listings with more amenities:
 - Move into higher price ranges (~ \$130 - \$250+)
- Additional features refine price further:
 - Having a view increases price significantly
 - Pet-friendly and other features adjust price within groups
- This shows:
 - Price depends on combinations of amenities
 - Not just one feature at a time

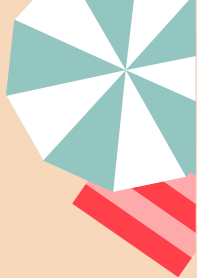




NLP on Reviews

- **VADER sentiment analysis**
- **TF-IDF keyword analysis**

Are there certain “key” words mentioned in the reviews, and does it correlate to higher reviews?



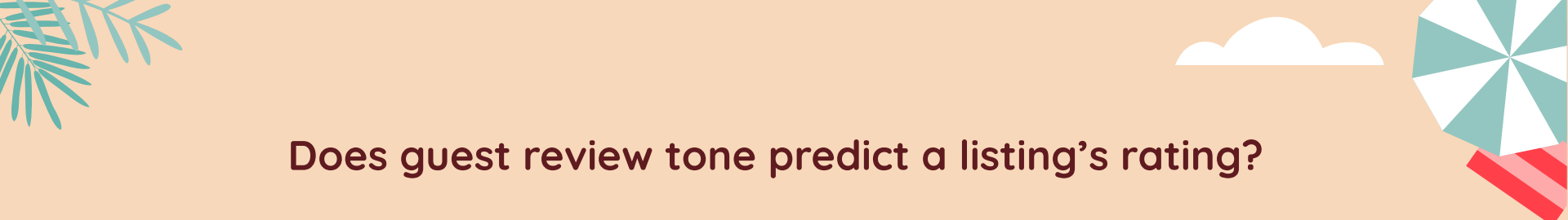
VADER Sentiment Analysis

Data Cleaning

```
#Run VADER on every review and compound score ranges from -1 (most negative) to +1 (most positive)  
reviews_clean['sentiment_score'] = reviews_clean['comments'].apply(  
    lambda x: sia.polarity_scores(x)['compound']  
)
```

Mean sentiment: 0.826
Min: -0.981 | Max: 0.999

→ aggregated sentiment scores to listing level, then imputed missing sentiment (61 rows) with median by room type



Does guest review tone predict a listing's rating?

Review Category	Equation	R ²
Mean Sentiment	$y = 1.03x + 4.02$	0.08 → Weak

Not really... NEXT

TF-IDF Section:

Split listings into high and low rated (4.9+ is high_rated and lower is low_rated)



TF-IDF Keyword Analysis

Keywords Associated with High vs Low Rated Listings

HIGH Rated Listings (≥ 4.9)

Keywords associated with High Rated Listings (≥ 4.9) include: wonderful, beautiful, room, space, snacks, comfortable, amenities, loved, garage, fresh, highly, recommend, neighborhood, bed, lovely, loft, absolutely, touches, time, and highly.

LOW Rated Listings (< 4.9)

Keywords associated with Low Rated Listings (< 4.9) include: location, house, family, creek, bathroom, bedroom, backyard, beds, kitchen, hosts, unit, cat, home, responsive, cottage, back, friendly, overall, and people.

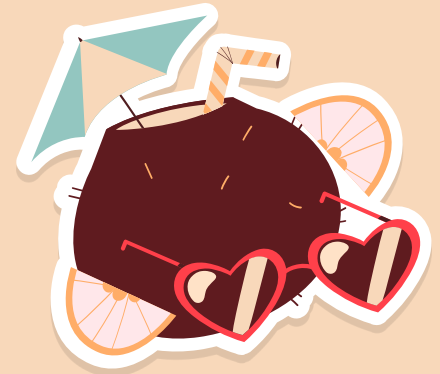
Takeaways: Hosts

- Add high-impact amenities and property characteristics to increase price and demand:
 - Parking, hot tubs, and views showed the biggest price increases
 - Historical features in homes tend to show higher prices
- Focus on “premium” features and personal touches, not just basics:
 - Wifi and AC are expected and won't justify higher pricing
 - Additional touches like snacks lead to higher ratings
- Increase total amenities strategically:
 - Listings with more amenities consistently price higher
- Keep your listing information accurate
- Stay responsive to guests throughout the entire stay, not just before check-in
- Stand out in a growing market:
 - More listings → stronger need to differentiate your property



Takeaways: Customers

- Choose properties that have a high accuracy rate
- Book earlier for summer stays in Salem
- Don't assume higher price = better experience:
 - Ratings are already high across most listings
 - Reading reviews can give valuable insights
- Pay for amenities you actually care about:
 - Higher prices mainly reflect the features, not necessarily better reviews
- More amenities does not always mean a better stay:
 - Consider what is most important to you in your stay



Future Work & Explorations



Future Features to Explore:

- Bedrooms
- Bathrooms
- Availability in coming months
- Proximity to landmarks
- Minimum stay time
- Maximum stay time
- Number of reviews

Potential Model:

Hosts

- Would input their amenities and features and get suggestions on what to prioritize in their listing and advertising, or maybe some features to implement

Guests

- Would input their priorities for what they want in an airbnb and our model would suggest some properties that best meet their needs





THANKS!

Questions?